

CHELTENHAM GIRLS HIGH SCHOOL



YEAR 12 COMMON TEST 2

2007

EXTENSION 2

Time allowed : 120 MINUTES

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- ALL questions are of equal value.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work
- Board approved calculators may be used.
- Each question is to be started on a **new page** and you are to write your name and teachers name on each page.
- The marks allocated for each question are indicated

Name : _____ Student Number: _____

Q	1	2	3	4	Total
Mark	/20	/20	/20	/20	/80

Question 1

- a) For the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$:
- i) determine the value of e (1)
 - ii) find the coordinates of the foci (1)
 - iii) find the equations of the directrices (1)
 - iv) sketch the ellipse (1)

b) Solve the equation $x^2 + 2x + 11 = 0$. (3)

- c) The polynomial $P(x) = x^4 - x^3 - 7x^2 + 13x - 6$ has a root of multiplicity 2.
- i) Show that 2 is a zero of $P(x)$. (1)
 - ii) Determine the multiple root. (2)
 - iii) Factorise $P(x)$ into linear factors. (1)

- d) From what external point is the tangent to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ to be drawn so that $4x - 3y + 12 = 0$ is the chord of contact.

You may assume that the chord of contact is from the external point

$$T(x_0, y_0) \text{ is } \frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1. \quad (2)$$

- e) For the polynomial $P(x) = x^3 - 2x^2 + x - 1$ determine the values of:

- i) $\alpha + \beta + \gamma$ (1)
- ii) $\alpha\beta\gamma$ (1)
- iii) $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$ (1)

- f) i) Write $z = 1 + i\sqrt{3}$ in modulus argument form. (2)

- ii) Hence determine the value of z^{10} . Give your answer in $x + iy$ form. (2)

Question 2

a) i) Divide the polynomial $f(x) = 2x^4 - 10x^3 + 12x^2 + 2x - 3$ by $g(x) = x^2 - 3x + 1$. (3)

ii) Hence write $f(x) = g(x)q(x) + r(x)$ where $q(x)$ and $r(x)$ are polynomials. (1)

iii) Hence show that $f(x)$ and $g(x)$ have no zeros in common. (1)

b) Write $\frac{4x^2 - 5x - 7}{(x-1)(x^2 + x + 2)}$ in the form $\frac{A}{x-1} + \frac{Bx + C}{x^2 + x + 2}$. (3)

c) The points P, Q and R represent the complex numbers w_1, w_2 and w_3 where $w_2 - w_1 = i(w_3 - w_1)$.

What kind of triangle is ΔPQR ? Explain your answer. (2)

d) Show that the condition for $y = mx + c$ to touch $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $c^2 = b^2 + a^2m^2$. (4)

e) Sketch the locus of z given that $z^2 - \bar{z}^2 = 16i$. (3)

f) Find the equation of the tangent to $2x^2 - y^3 = 1$ at $x = -1$. (3)

Question 3

- a) i) Find the real numbers a and b such that

$$x^4 + x^3 + x^2 + x + 1 = (x^2 + ax + 1)(x^2 + bx + 1). \quad (3)$$

- ii) Find the solutions of $x^4 + x^3 + x^2 + x + 1 = 0$. (3)

- iii) Hence determine the exact value of $\cos \frac{2\pi}{5}$. (2)

- b) i) Show that $P(a \cos \theta, b \sin \theta)$ lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. (1)

- ii) S and S' are the foci of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Show that $SP = a - ae \cos \theta$. (3)

- iii) If $SP' = a + ae \cos \theta$ show that $SP + S'P = 2a$ (1)

- c) i) Show that the equation of the normal at $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\text{is } \frac{(x - x_1)y_1}{b^2} = \frac{(y - y_1)x_1}{a^2}. \quad (4)$$

- ii) Show that the only normal that passes through a focus is the x - axis. (3)

Question 4

a) The roots of $x^3 + qx + r = 0$ are α, β and γ .

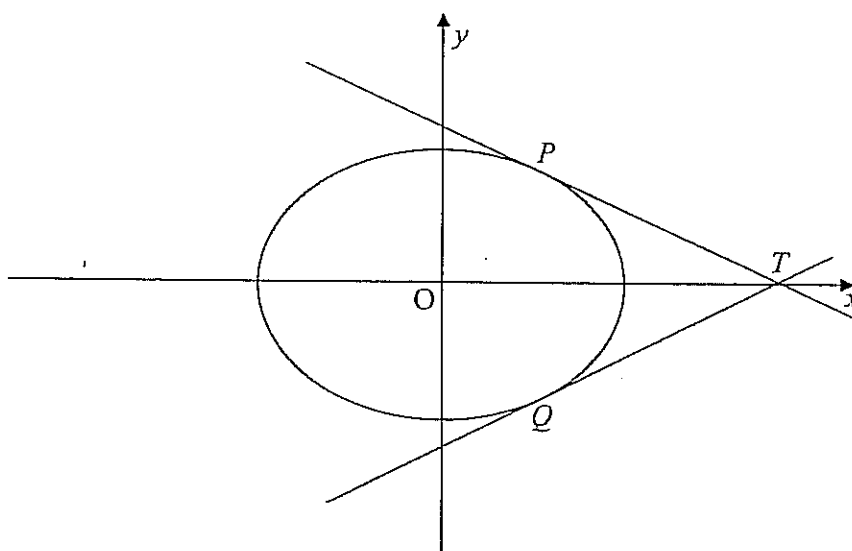
i) Determine $\alpha + \beta + \gamma$. (1)

ii) Determine $\alpha\beta + \beta\gamma + \alpha\gamma$. (1)

iii) Determine $\alpha\beta\gamma$. (1)

iv) Hence prove $(\beta - \gamma)^2 + (\gamma - \alpha)^2 + (\alpha - \beta)^2 = -6q$. (4)

b) The point P lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The tangents to the ellipse through P and Q meet at the point $T(\frac{a}{e}, 0)$.



i) Determine the equation of the chord of contact. (4)

ii) What is the value of the ratio $\frac{PS}{ST}$? (1)

iii) Show that $\angle PTQ$ is less than a right angle. (1)

iv) Show that the area of ΔPTQ is $\frac{b^4}{a^2 e}$. (3)

Question 4 is continued on the next page.

Question 4 (continued)

c) Let $w = \frac{3+4i}{5}$ and $z = \frac{5+12i}{13}$, so that $|w| = |z| = 1$. Do not prove this.

i) Find wz and $w\bar{z}$ in the form $x + iy$. (2)

ii) Hence find 2 distinct ways of writing 65^2 as the sum of a^2 and b^2 , where a and b are integers and $0 < a < b$. (2)

END OF EXAMINATION

Question 1

$$a) \quad \frac{x^2}{16} + \frac{y^2}{4} = 1$$

$$i) \quad b^2 = a^2(1 - e^2)$$

$$4 = 16(1 - e^2)$$

$$\frac{4}{16} = 1 - e^2$$

$$e^2 = 1 - \frac{4}{16}$$

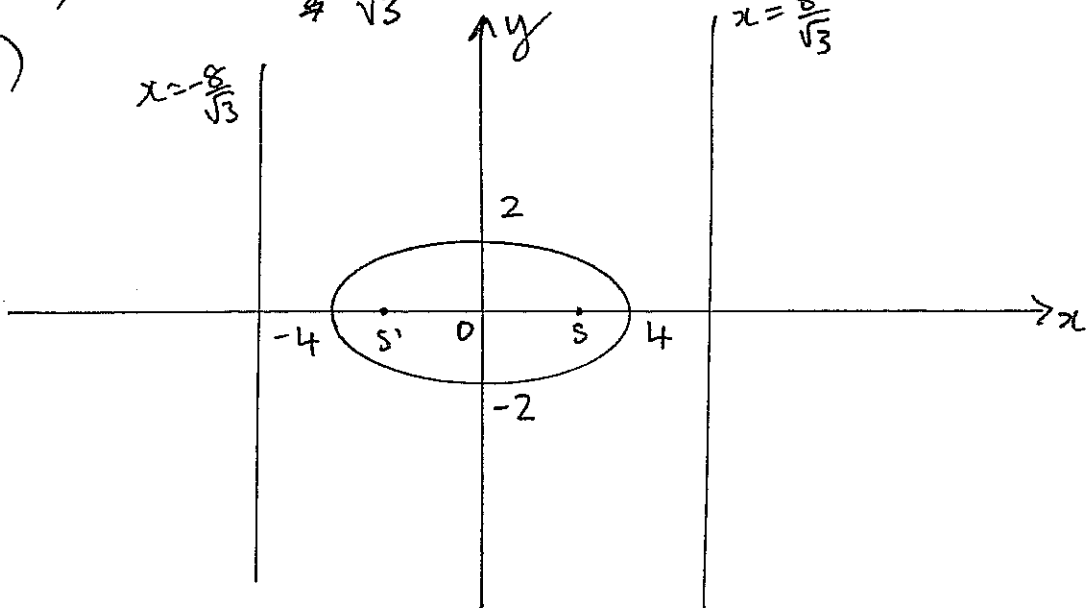
$$e = \frac{\sqrt{3}}{2}$$

$$ii) \quad S(2\sqrt{3}, 0) \quad S'(-2\sqrt{3}, 0)$$

$$iii) \quad x = \frac{8}{\sqrt{3}}$$

iv)

$$x = -\frac{8}{\sqrt{3}}$$



$$b) \quad x^2 + 2x + 11 = 0$$

$$x = \frac{-2 \pm \sqrt{4 - 4 \cdot 1 \cdot 11}}{2}$$

$$= \frac{-2 \pm \sqrt{-40}}{2}$$

$$= \frac{-2 \pm 2i\sqrt{10}}{2} = -1 \pm i\sqrt{10}$$

$$c) \ i) \ P(2) = 2^4 - 2^3 - 28 + 26 - 6 \\ = 0$$

$\therefore x = 2$ is a zero.

$$ii) \ P'(x) = 4x^3 - 3x^2 - 14x + 13$$

$$P'(1) = 4 - 3 - 14 + 13 = 0$$

$$P(1) = 0$$

$\therefore x = 1$ is a repeated root.

$(x-1)^2(x-2)$ are factors of $P(x)$

$$(x^2 - 2x + 1)(x-2) = x^3 - 2x^2 - 2x^2 + 4x + x - 2 \\ = x^3 - 4x^2 + 5x - 2$$

$$\begin{array}{r} x^3 - 4x^2 + 5x - 2 \overline{) x^4 - x^3 - 7x^2 + 13x - 6} \quad x-3 \\ \underline{x^4 - 4x^3 + 5x^2 - 2x} \\ 3x^3 - 12x^2 + 15x - 6 \\ \underline{3x^3 - 12x^2 + 15x - 6} \\ 0 \end{array}$$

$$iii) \ P(x) = (x-1)^2(x-2)(x-3).$$

$$d) \text{ chord of contact : } \frac{xx_0}{9} + \frac{yy_0}{4} = 1$$

$$\Rightarrow 4xx_0 + 9yy_0 = 36$$

$$\text{eqn. of chord : } 4x_0 - 3y_0 = -12 \Rightarrow -12x + 9y = 36$$

$$\text{equating coefficients : } 4x_0 = -12 \quad ; \quad 9y_0 = 36 \\ x_0 = -3 \quad \quad \quad y_0 = 1$$

the external pt is $(-3, 1)$

$$e) \alpha + \beta + \gamma = 2$$

$$\alpha\beta\gamma = 1$$

$$\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma} = \frac{\alpha + \beta + \gamma}{(\alpha\beta\gamma)^2}$$

$$= \frac{2}{1}$$

$$= 2$$

Question 2.

a) i)

$$\begin{array}{r} x^2 - 3x + 1 \overline{) 2x^4 - 10x^3 + 12x^2 + 2x - 3} \\ \underline{2x^4 - 6x^3 + 2x^2} \\ -4x^3 + 10x^2 + 2x \\ \underline{-4x^3 + 12x^2 - 4x} \\ -2x^2 + 6x - 3 \\ \underline{-2x^2 + 6x - 2} \\ -1 \end{array}$$

$$ii) f(x) = (x^2 - 3x + 1)(2x^2 - 4x - 2) - 1$$

iii) if they had zeros in common $\Rightarrow$ they had common factors
 $\Rightarrow g(x)$ was a factor of $f(x)$.

$$b) \frac{4x^2 - 5x - 7}{(x-1)(x^2+x+2)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+x+2}$$

$$4x^2 - 5x - 7 = A(x^2+x+2) + (Bx+C)(x-1)$$

$$\text{when } x=1, \quad 4-5-7 = 4A$$

$$-8 = 4A$$

$$A = -2$$

$$\text{when } x=0, \quad -7 = 2A - C$$

$$-7 = -4 - C$$

$$C = 3$$

$$\text{when } x=-1, \quad 4+5-7 = 2A + (-B+C)(-2)$$

$$2 = 2A - 2B + 2C$$

$$4 + 2 + 6 = 2B$$

$$B = 6$$

c) a right isosceles $\therefore \left| \frac{w_2 - w_1}{w_3 - w_1} \right| = |i| = 1$ Δ isosceles $\therefore |w_2 - w_1| = |w_3 - w_1|$ $\therefore x \text{ by } i = \text{as a } \pi/2$ by right angled

$$d). \quad \frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} = 1$$

$$b^2 x^2 + a^2 (m^2 x^2 + 2mx + c^2) = a^2 b^2$$

$$b^2 x^2 + a^2 m^2 x^2 + 2m x a^2 + c^2 a^2 - a^2 b^2 = 0$$

$$\Delta = 0$$

$$= (2m a^2)^2 - 4 \cdot (b^2 + a^2 m^2) (a^2 c^2 - a^2 b^2)$$

$$= 4m^2 a^4 - 4(b^2 + a^2 m^2)(a^2 c^2 - a^2 b^2)$$

$$\frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} = 1$$

$$b^2 x^2 + a^2 (m^2 x^2 + 2mcx + c^2) = a^2 b^2$$

$$b^2 x^2 + a^2 m^2 x^2 + 2a^2 mcx + a^2 c^2 - a^2 b^2 = 0$$

$$(b^2 + a^2 m^2) x^2 + 2a^2 mcx + (a^2 c^2 - a^2 b^2) = 0$$

$$\Delta = 0$$

$$0 = (2a^2 mc)^2 - 4(b^2 + a^2 m^2)(a^2 c^2 - a^2 b^2)$$

$$= 4a^4 m^2 c^2 - 4(a^2 b^2 c^2 - a^2 b^4 + a^4 m^2 c^2 - a^4 m^2 b^2)$$

$$0 = 4a^4 m^2 c^2 - 4a^2 b^2 c^2 + 4a^2 b^4 - 4a^4 m^2 c^2 + 4a^4 m^2 b^2$$

$$0 = -4a^2 b^2 c^2 + 4a^2 b^4 + 4a^4 m^2 b^2$$

$$4a^2 b^2 c^2 = 4a^2 b^4 + 4a^4 m^2 b^2$$

$$c^2 = b^2 + a^2 m^2$$

e) $z^2 - \bar{z}^2 = 16i$

$$(x+iy)^2 = x^2 + 2xyi - y^2$$

$$= x^2 - y^2 + 2xyi$$

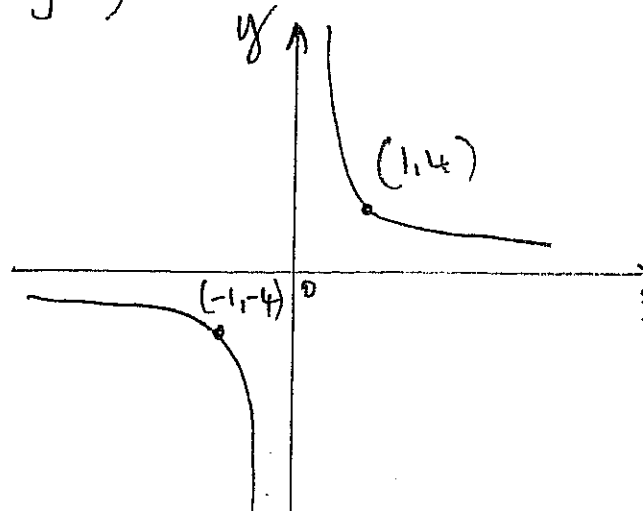
$$\bar{z}^2 = x^2 - y^2 - 2xyi$$

$$x^2 - y^2 + 2xyi - (x^2 - y^2 - 2xyi) = 16i$$

$$4xyi = 16i$$

$$4xy = 16$$

$$xy = 4$$



$$Q2 f). \quad 2x^2 - y^3 = 1 \quad \text{at } x = -1 \quad \Rightarrow \quad 2 - y^3 = 1$$

$$4x - 3y^2 \cdot \frac{dy}{dx} = 0$$

$$1 = y^3$$

$$y = 1$$

$$4x = 3y^2 \cdot \frac{dy}{dx}$$

$$\frac{4x}{3y^2} = \frac{dy}{dx}$$

$$\text{at } x = -1, \quad \frac{dy}{dx} = -\frac{4}{3}$$

$$-\frac{4}{3} = \frac{y-1}{x+1}$$

$$-4x - 4 = 3y - 3$$

$$4x + 3y + 1 = 0$$

Question 3

$$a)(x^2 + ax + 1)(x^2 + bx + 1)$$

$$= x^4 + bx^3 + x^2 + ax^3 + abx^2 + ax + x^2 + bx + 1$$

$$= x^4 + (a+b)x^3 + (2+ab)x^2 + (a+b)x + 1$$

$$a+b = 1, \quad 2+ab = 1$$

$$ab = -1$$

$$b = -\frac{1}{a}$$

$$a - \frac{1}{a} = 1$$

$$a^2 - a = 1$$

$$a^2 - a - 1 = 0$$

$$a = \frac{1 \pm \sqrt{1+4 \cdot 1 \cdot 1}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

$$b = -\frac{1}{\frac{1 \pm \sqrt{5}}{2}} \rightarrow$$

$$\text{if } a = \frac{1+\sqrt{5}}{2}$$

$$-1 \div \frac{1+\sqrt{5}}{2}$$

$$= -1 \times \frac{2}{1+\sqrt{5}} \cdot \frac{1-\sqrt{5}}{1-\sqrt{5}}$$

$$= \frac{-2+2\sqrt{5}}{1-5}$$

$$= \frac{-1+\sqrt{5}}{-2}$$

$$= \frac{1-\sqrt{5}}{2}$$

$$a = \frac{1+\sqrt{5}}{2}, \quad b = \frac{1-\sqrt{5}}{2}$$

ii)

$$x^5 = 1$$

$$x^5 = \cos 2\pi k$$

$$x = \cos \frac{2\pi k}{5}$$

$$x_0 = \cos 0$$

$$x_1 = \cos \frac{2\pi}{5}$$

$$x_2 = \cos \frac{4\pi}{5}$$

$$x_3 = \cos \frac{6\pi}{5} = \cos\left(-\frac{4\pi}{5}\right)$$

$$x_4 = \cos \frac{8\pi}{5} = \cos\left(-\frac{2\pi}{5}\right)$$

sols of

$$x^4 + x^3 + x^2 + x + 1 = 0$$

are

$$\cos\left(\pm \frac{2\pi}{5}\right), \cos\left(\pm \frac{4\pi}{5}\right)$$

$\left(x - \cos \frac{2\pi}{5}\right)\left(x - \cos\left(-\frac{2\pi}{5}\right)\right)$ is a quadratic factor

$$= x^2 - 2\cos \frac{2\pi}{5} x +$$

iii) so equating coefficients with

$$x^2 + \frac{b}{a}x + 1, \quad \cos \frac{2\pi}{5} > 0$$

$$-2\cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{2}$$

$$\cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4}$$

$$b i) \quad \frac{(a \cos \theta)^2}{a^2} + \frac{(b \sin \theta)^2}{b^2}$$

$$= \frac{a^2 \cos^2 \theta}{a^2} + \frac{b^2 \sin^2 \theta}{b^2}$$

$$= \cos^2 \theta + \sin^2 \theta$$

$$= 1$$

= R.H.S

$\therefore$ P lies on ellipse

$$ii) \quad S(ae, 0), \quad P(a \cos \theta, b \sin \theta)$$

$$SP = \sqrt{(ae - a \cos \theta)^2 + (0 - b \sin \theta)^2}$$

$$= \sqrt{a^2 e^2 - 2a^2 e \cos \theta + a^2 \cos^2 \theta + b^2 \sin^2 \theta}$$

$$= \sqrt{a^2 e^2 - 2a^2 e \cos \theta + a^2 \cos^2 \theta + b^2 - b^2 \cos^2 \theta}$$

$$= \sqrt{a^2 - b^2 - 2a^2 e \cos \theta + b^2 + (a^2 - b^2) \cos^2 \theta}$$

$$= \sqrt{a^2 - 2a^2 e \cos \theta + a^2 e^2 \cos^2 \theta}$$

$$= \sqrt{(a - ae \cos \theta)^2}$$

$$= a - ae \cos \theta$$

$$iii) \quad SP' = a + ae \cos \theta \quad \therefore \quad SP + SP' = a + ae \cos \theta + a - ae \cos \theta = 2a$$

$$c) i) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{2y}{b^2} \cdot \frac{dy}{dx} = -\frac{2x}{a^2}$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y}$$

$$= -\frac{b^2 x}{a^2 y}$$

at $P(x_1, y_1)$, $\frac{dy}{dx} = -\frac{b^2 x_1}{a^2 y_1}$ the gradient of normal is $\frac{a^2 y_1}{b^2 x_1}$

$$\frac{y - y_1}{x - x_1} = \frac{a^2 y_1}{b^2 x_1}$$

$$b^2 x_1 y - b^2 x_1 y_1 = a^2 x y_1 - a^2 x_1 y_1$$

$$b^2 x_1 y - b^2 x_1 y_1 = a^2 (x - x_1) y_1$$

$$b^2 (y - y_1) x_1 = a^2 (x - x_1) y_1$$

$$\frac{(y - y_1) x_1}{a^2} = \frac{(x - x_1) y_1}{b^2}$$

ii) if this passes through $(ae, 0)$

then $-\frac{y_1 x_1}{a^2} = \frac{(ae - x_1) y_1}{b^2}$, $\div y_1$, assumes that $y_1 \neq 0$

$$-\frac{x_1}{a^2} = \frac{ae - x_1}{b^2}$$

$$-\frac{b^2}{a^2} x_1 = ae - x_1$$

$$x_1 - \frac{b^2}{a^2} x_1 = ae$$

$$x_1 \left(1 - \frac{b^2}{a^2}\right) = ae$$

$$x_1 e^2 = ae$$

$$x_1 = \frac{a}{e}$$

but x_1 is on the ellipse not the

Question 4

a) i) $\alpha + \beta + \gamma = 0$

ii) $\alpha\beta + \beta\gamma + \alpha\gamma = 9$

iii) $\alpha\beta\gamma = -1$

iv)

$$\beta + \gamma = -\alpha$$

$$(\beta - \gamma)^2 = \beta^2 - 2\beta\gamma + \gamma^2$$

$$= \beta^2 + \gamma^2 - 2\beta\gamma$$

$$= (\beta + \gamma)^2 - 4\beta\gamma$$

$$= \alpha^2 - 4\beta\gamma$$

$$(\beta - \gamma)^2 + (\gamma - \alpha)^2 + (\alpha - \beta)^2$$

$$= \alpha^2 - 4\beta\gamma + \beta^2 - 4\gamma\alpha + \gamma^2 - 4\alpha\beta$$

$$= \alpha^2 + \beta^2 + \gamma^2 - 4(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= (\alpha + \beta + \gamma)^2 - 6(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= 0^2 - 6 \cdot 9$$

$$= -54$$

$$(\alpha + \beta + \gamma)^2$$

$$= (\alpha + \beta)^2 + 2(\alpha + \beta)\gamma$$

$$= \alpha^2 + 2\alpha\beta + \beta^2 + 2\alpha\gamma + 2\beta\gamma + \gamma^2$$

b) eqn of tangent at $P(x_1, y_1)$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y}$$

$$= -\frac{b^2 x}{a^2 y}$$

$$\text{at } P, \frac{dy}{dx} = -\frac{b^2 x_1}{a^2 y_1}$$

eqn of PT

$$\frac{b^2 x_1}{a^2 y_1} = \frac{y}{x - \frac{a}{e}}$$

$$b^2 x x_1 - b^2 \frac{a}{e} x_1 = y y_1 a^2$$

$$a^2 y y_1 + b^2 x x_1 = \frac{b^2 a}{e} x_1$$

$$\frac{x x_1}{a^2} + \frac{y y_1}{b^2} = \frac{b^2 a}{e a^2 b^2} x_1$$

$$\frac{x x_1}{a^2} + \frac{y y_1}{b^2} = \frac{x_1}{a e}$$

similarly eqn of QT is

$$\frac{x x_2}{a^2} + \frac{y y_2}{b^2} = \frac{x_2}{a e}$$

chord of contact

$$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = \frac{x_1}{ae}$$

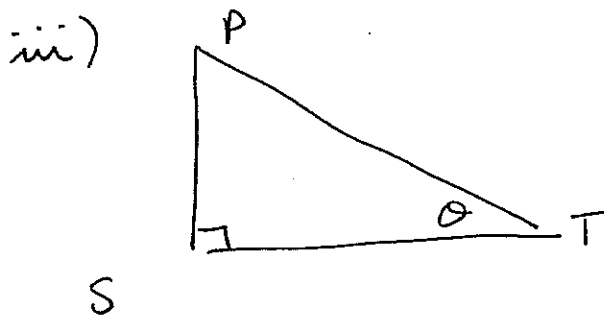
$$\frac{x \cdot \frac{a}{e}}{a^2} = \frac{x_1}{ae}$$

$$x = \frac{x_1}{ae} \times \frac{a^2}{a/e}$$

$$= \frac{x_1 a^2}{a^2}$$

$$x = x_1$$

$$\text{ii)} \quad \frac{PS}{ST} = \frac{PS}{PM} = e$$



$$\tan \theta = \frac{PS}{ST} = e, \quad e < 1$$

$$\tan \theta < 1$$

$$\theta < \frac{\pi}{4}$$

$$\angle PTQ = 2\theta$$

$$\angle PTQ < \frac{\pi}{2}$$

$$\begin{aligned} \text{iv)} \quad \text{area} &= \frac{1}{2} \times PQ \times ST \\ &= \frac{1}{2} \times 2e \times ST \times ST \end{aligned} \quad \begin{aligned} 2PS &= PQ \\ PS &= e ST \end{aligned}$$

$$= \frac{1}{2} \times 2e \times \left(\frac{a}{e} - ae \right)^2$$

$$= e \left(\frac{a - ae^2}{e} \right)^2$$

$$= e \left(a \left(\frac{1}{e} - e \right) \right)^2$$

$$= ea^2 \left(\frac{1}{e} - e \right)^2$$

$$= ea^2 \left(\frac{1 - e^2}{e} \right)^2$$

$$= \frac{ea^2}{e^2} (1 - e^2)^2$$

$$= \frac{ea^2}{e^2} \cdot \left(\frac{b^2}{a^2} \right)^2$$

$$= \frac{b^4}{a^2 e}$$

$$b) \quad w = \frac{3+4i}{5}, \quad z = \frac{5+12i}{13}$$

$$wz = \left(\frac{3+4i}{5} \right) \left(\frac{5+12i}{13} \right)$$

$$= \frac{15 + 36i + 20i - 48}{65}$$

$$= \frac{-33 + 56i}{65}$$

$$w\bar{z} = \left(\frac{3+4i}{5} \right) \left(\frac{5-12i}{13} \right)$$

$$= \frac{15 - 36i + 20i + 48}{65}$$

$$= \frac{63 - 16i}{65}$$

$$ii) \quad |wz| = \sqrt{\left(\frac{-33}{65}\right)^2 + \left(\frac{56}{65}\right)^2} = 1 \quad \text{as} \quad |w| = |z| = 1$$

$$\Rightarrow 33^2 + 56^2 = 65^2$$

$$|w\bar{z}| = \sqrt{\left(\frac{63}{65}\right)^2 + \left(\frac{-16}{65}\right)^2} = 1 \quad \text{as} \quad |w| = |\bar{z}| = 1$$

$$|z| = |\bar{z}| = 1$$

$$\Rightarrow 63^2 + 16^2 = 65^2$$

